

“TiDB supports scaling to millions of queries per second”

The inverse correlation between the need for data storage and the rate at which data is processed has led to companies searching for database management systems that can keep up with the rapid growth of data. In conversation with *OSFY's Yashasvini Razdan, Bhanu Jamwal*, Head of Presales and Solution Engineering, APAC, TiDB, explains how distributed open source databases like TiDB are an alternative for organisations dealing with massive and complex data requirements.

Bhanu Jamwal, Head of Presales and Solution Engineering, APAC, TiDB



Q. What is the need for another database like TiDB when MySQL and PostgreSQL exist?

A. Traditional databases like MySQL and PostgreSQL were initially designed for smaller datasets and simpler workloads. While they serve well for small-scale operations, they struggle with scalability and performance as data sizes grow or query demands increase. This creates bottlenecks for organisations experiencing rapid growth. For instance, as businesses adopt SaaS (software-as-a-service) models, there is a need to handle not just more data but also higher query volumes and more database tables, all while maintaining performance. Traditional systems often cannot scale horizontally to meet these demands.

TiDB is a distributed database specifically designed to eliminate bottlenecks by offering scalability. Businesses can start small and grow as needed without compromising performance. TiDB supports scaling to millions of queries per second, which is essential for high-traffic applications. It can handle up to one million tables

on a single cluster, making it an ideal solution for SaaS companies onboarding multiple tenants.

Q. How does distributed architecture impact scalability and performance?

A. Distributed databases like TiDB ensure that regardless of data size or performance expectations, the system remains consistent. For instance, if your data grows from 1 terabyte to 100 terabytes, TiDB will deliver the same performance levels. This is achieved by scaling out the cluster—adding more computational resources or storage as needed. As a result, there is no performance degradation, ensuring the applications remain unaffected, regardless of growth.

Q. How does TiDB compete with proprietary distributed databases or cloud-native offerings like Google Spanner and AWS Aurora?

A. TiDB is open source, which offers two key advantages — access to ongoing innovation and freedom from vendor lock-in. Organisations can

rely on TiDB's open source nature for continuity and flexibility, regardless of changes in the industry.

It stands out from other distributed databases because of its hybrid transactional and analytical processing (HTAP) capabilities. Unlike many databases optimised solely for online transaction processing (OLTP), TiDB can handle both transactional and analytical workloads on the same cluster. This eliminates the need for ETL (extract, transform, load) processes, enabling real-time analytics. For instance, organisations can detect anomalies or fraud in real time, which is not possible with traditional databases or even some distributed ones. This HTAP capability is critical for companies needing real-time insights without added complexity.

Another differentiator is its compatibility with MySQL, with the ability to run transactional and analytical workloads on the same database, which makes it easy for organisations dependent on MySQL to migrate to TiDB without significant disruption—a simple lift-and-shift process.